1. IDENTIFICATION OF THE MIXTURE AND OF THE COMPANY

1.1 Product identifier

Trade name
Clomate

1.2 Relevant identified uses of the mixture and uses advised against

1.2.1 Relevant identified uses
A residual herbicide for the control of a range of annual weeds in oilseed rape, potato, carrot, vining pea, combining pea (spring), field bean (winter) and field bean (spring).

1.2.2 Uses advised against
Do not use for any other purpose.

1.3 Details of the supplier of the safety data sheet

Albaugh Europe Sàrl
World Trade Center Lausanne
Avenue Gratta-Paille 2
1018 Lausanne, Switzerland

Telephone:	+ 41 21 799 9130
Fax:	+ 41 21 799 9139
Email:	sds@albaugh.eu
Web:	www.albaugh.eu

1.4 Emergency telephone number

For advice on medical emergencies, fires or major spills: +44 (0) 1235 239 670

Available:	24 h
Time Zone:	GMT
Language(s) of phone service:	All major EU languages

Při ohrožení života a zdraví (Česká republika): Toxikologické informační středisko (TIS),
Klinika nemocí z povolání, Na bojišti 1, 128 08 Praha 2, (+420) 224 919 293 nebo (+420) 224 915 402

Available:	24h
Time Zone:	GMT
Language(s) of phone service:	Czech

2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) no. 1272/2008 [CLP/GHS]

Signal word	Hazard class and category	Pictograms	Hazard statement	M-Factor
-	Aquatic Chronic 3	-	H412 Harmful to aquatic life with long lasting effects	-

Additional information
For abbreviations, refer to Section 16.

2.2 Label elements

Labelling according to Regulation (EC) no. 1272/2008

Hazard pictograms
-

Signal word:
-

Hazard statements:
H412: Harmful to aquatic life with long lasting effects

Precautionary statements:
General P102: Keep out of reach of children.

Prevention: P270: Do not eat, drink or smoke when using this product.
P280: Wear protective gloves/protective clothing/eye protection/face protection.
Response: -
Storage: -
Disposal: P501: Dispose of contents/container according to local regulations

Supplemental information:
EUH401: To avoid risks to human health and the environment, comply with the instructions for use.
SP1: Do not contaminate water with the product or its container (Do not clean application equipment near surface water/Avoid contamination via drains from farmyards and roads).
The product is excluded from use in surface water protection zone II.
The product requires no special measures for the protection of birds, terrestrial vertebrates, bees, non-target arthropods, soil macro and micro-organisms.
Be sure to read the enclosed instructions!
Safety Data Sheets are available on request for professional users.

2.3 Other hazards
None known

3. COMPOSITION / INFORMATION ON INGREDIENTS

3.2 Mixtures
Description of the mixture:
Mixture of clomazone and co-formulants.

Chemical Name	CAS-No.	EC-No.	Index No.	Concentration (W/W)	CLP (Reg. 1278/2008) Classification
Clomazone	81777-89-1	-	-	30.3 %	Acute Tox. 4 (oral and inhalation); H302, H332. Aquatic Acute 1; H400 Aquatic Chronic 1; H410
Calcium chloride	10043-52-4	233-140-8	017-013-00-2	>5.0 <10.0 %	Eye Irrit. 2; H319
Other ingredients				to 100%	Not classified

Additional information
For full text of H phrases, see Section 16.

4. FIRST AID MEASURES

4.1 Description of first aid measures
General notes:
If symptoms occur after exposure to this product, seek medical attention immediately and show the product label or this SDS. Remove to fresh air and keep at rest. Do not allow smoking or eating. Take off all contaminated clothing and footwear.
Following inhalation:
Remove to fresh air and keep at rest in half-upright position. Seek medical attention if symptoms arise.
Following skin contact:
Remove all contaminated clothing. Wash skin with soap and rinse with plenty of water. Seek medical attention if irritation arises. Wash clothes before re-use.
Following eye contact:
Immediately rinse with water. Holding eyes open, continue rinsing for 15 minutes at least. Remove contact lenses as soon as possible. Seek medical attention if irritation arises.
Following ingestion:
If swallowed, DO NOT INDUCE VOMITING: seek medical advice immediately and show this container or label. Remove any residues from mouth and rinse it with plenty of water. Offer the casualty 1 or 2 glasses of water to drink. Never give anything by mouth to an unconscious person.
Self-protection of first aider
Personal protective equipment for first aid responders is recommended according to potential for exposure (refer to Section 8).

4.2 Most important symptoms and effects, both acute and delayed
The symptoms and the effects indicated in this section refer to an accidental exposure scenario.

Following inhalation:
Possible slight nasal irritation and discharge. No delayed effects expected.
Following skin contact:
Possible slight transitory redness. No delayed effects expected.
Following eye contact:
Possible slight transitory redness. No delayed effects expected.
Following ingestion:
Possible mild gastrointestinal effects. No significant delayed effects expected.

4.3 Indication of immediate medical attention and special treatment needed
No need to provide any special means/medicinal products for immediate treatment at the workplace
Notes for the doctor:
No specific antidote. Treat symptomatically (decontamination, vital functions). Call a Poison Centre immediately for treatment advice. In case of ingestion gastric lavage may be necessary (with proper laryngeal control). Before emptying the stomach, assess the potential danger arising from lung aspiration against the product toxicity. Report to Albaugh Europe Sàrl any unusual symptoms occurring after exposure by any route.

5. FIRE FIGHTING MEASURES

5.1 Extinguishing media
Suitable extinguishing media:
Carbon dioxide, water spray, alcohol-resistant foam, dry chemical for small fires, alcohol-resistant foam or water spray for large fires.
Unsuitable extinguishing media:
Solid water jet.

5.2 Special hazards arising from the mixture
Hazardous combustion products
Evolves toxic and irritant fumes in fire including nitrogen oxides and chlorine oxides.

5.3 Advice for fire-fighters
Clothing conforming to EN469 should be sufficient to deal with fires involving the substance.
However, a Self-Contained Breathing Apparatus (SCBA) may be required if there is a potential for exposure to combustion fumes.

5.4 Additional information
Provide storage and work areas with suitable fire extinguishers.
Call the Fire Brigade at once to deal with all fires involving pesticides unless the fire is small and immediately controllable. Spray unopened containers with a mist spray to keep cool. If without risk, remove intact containers from exposure to fire. Contain fire-fighting water, bunding if necessary with sand or earth. Do not allow contamination of public drains or surface or ground waters. Dispose of fire debris and contaminated water according to all relevant national legislation.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures
6.1.1 For non-emergency personnel
Protective equipment: Remove immediately any contaminated clothing. Wear prescribed personal protective equipment to prevent contact with eyes and skin. A Self-Contained Breathing Apparatus (SCBA) may be required if there is an elevated risk for exposure.
Emergency procedures: Call the emergency services if the release is not immediately controllable. If the release is localized and immediately controllable, provide sufficient ventilation and control the release at its source.
6.1.2 For emergency responders
Clothing conforming to EN469.

6.2 Environmental precautions
Use appropriate containment to avoid environmental contamination. Control the release at its source. Contain spills to prevent it from spreading, contaminating soil or entering sewage and drainage systems or any body of water. Inform the local water authorities if the release enters drains, surface or ground waters.

6.3 Methods and material for containment and cleaning up

For containment
Clean up spills immediately and place in a compatible disposal container. Contain spill by diking with earth, sand or absorbent material and place into a compatible marked disposal container.
For cleaning up
Scrub area with a hard water detergent. Soak up wash liquid with additional absorbent material and place into a compatible marked disposal container. Seal container and arrange for disposal.
Other information
Not Applicable

6.4 Reference to other sections
Refer to Section 8 for personal protective equipment and to Section 13 for disposal instructions.

7. HANDLING AND STORAGE

7.1 Precautions for safe handling
Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Provide suitable ventilation in the areas where the product is stored and used. Contaminated work clothing should not be allowed out of the workplace. Avoid all contact by mouth, with eyes and skin. Wear personal protective equipment as specified in Section 8. When using, do not eat, drink or smoke. Remove contaminated clothing and protective equipment before meals and after work. Wash hands and exposed skin before meals and after work. Wash all protective clothing thoroughly after use, especially the insides of gloves.

7.2 Conditions for safe storage, including any incompatibilities
The material is stable under normal ambient conditions. Keep in original container, in a dry, cool and safe place. Store in a locked, suitable store. Keep away from any source of ignition. Keep out of the reach of children and unauthorised personnel. Keep away from food, drink and animal feeding stuffs.

7.3 Specific end use(s)
Product for professional use as directed by the product label, every other use is hazardous.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

8.1 Control Parameters
Occupational Exposure limit values
No Occupational Exposure limit values have been set for the mixture or its components:

Information on monitoring procedures
None available.

8.2 Exposure controls

8.2.1 Appropriate engineering controls
Engineering controls and appropriate work processes must be used to eliminate or reduce worker and environmental exposure in the areas where the substance is handled, transported, loaded, unloaded, stored and used. These measures must be adequate for the extent of the actual risk. Provide adequate local exhaust ventilation. Use specialized transfer systems if available.

8.2.2 Personal protection equipment
Eye and face protection
Avoid contact with eyes. If there is a significant potential for contact, wear suitable eye and face protection (EN 166).
Skin protection:
Hand protection: Wear suitable protective gloves against chemicals (EN 374 part 1, 2, 3). Nitrile rubber min. 0.5mm thick and 300mm long gloves are the ones proven to be the most suitable according to tests on pesticide products.
Wash the gloves thoroughly after each use, especially the insides. Replace gloves if damaged and before exceeding the breakthrough time.
Body protection: Avoid contact with skin. If there is a significant potential for contact, wear suitable coveralls (ISO 13982-1, Type 5, EN 13034, Type 6).
Other skin protection: None specified.
Respiratory protection: No special requirement when used as recommended. If a risk assessment shows that engineering controls do not provide adequate respiratory protection to exposure to spray particles, wear particle filtering half mask (EN 149) or half mask connected to particle filter (EN 140 + 143).

8.2.3 Environmental exposure controls

Implement all applicable local and community environmental protection legislation. Refer to Section 15. Use appropriate containment to avoid environmental contamination. Do not empty into drains. Do not contaminate water with the product or used container. Do not clean application equipment near surface water. Avoid contamination via drains from farmyards and roads. Refer to Section 12 and 13.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

All the data contained in this section are derived from actual test data on the mixture or its components unless otherwise stated.

a)	*Appearance:*	Liquid
	Colour:	Off-white
b)	*Odour:*	Faint amine
c)	*Odour threshold:*	Not determined – not required under all applicable pesticide legislation
d)	*pH:*	9.78 (1% dilution in water)
e)	*Melting point/freezing point:*	Not applicable – the mixture is a liquid at ambient temperature and must be protected from frost
f)	*Initial boiling point and boiling range:*	Approximately 100 °C
g)	*Flash point:*	None – the mixture is water-based
h)	*Evaporation rate:*	Not available - not required under all applicable pesticide legislation
i)	*Flammability (solid, gas):*	Not applicable (liquid)
j)	*Upper/lower flammability or explosive limits:*	Not applicable – the mixture is water-based
k)	*Vapour pressure:*	Not applicable to the mixture – not required under all applicable pesticide legislation Clomazone: 2.7×10^{-2} Pa
l)	*Vapour density:*	Not applicable – not required under all applicable pesticide legislation
m)	*Density:*	1.19 g/cm^3 at 20°C
n)	*Solubility(ies)*	
	Solubility (water):	Forms a stable dispersion in water
o)	*Partition coefficient: n-octanol/water:*	Clomazone: Kow 2.58 @ pH7, 20°C
p)	*Auto-ignition temperature:*	>400 °C
	Minimum Ignition Temperature:	Not available
	Minimum Ignition Energy:	Not available
q)	*Decomposition temperature:*	Clomazone: >281°C
r)	*Viscosity:*	158 mPa.s at 20°C/ 188 mPa.s at 40°C
s)	*Explosive properties:*	Explosion hazard: not explosive.
t)	*Oxidising properties:*	Not an oxidising agent.

9.2 Other information

Surface tension: 51.3 mNm^{-1}
Flammability (in contact with water): not flammable

10. STABILITY AND REACTIVITY

10.1 Reactivity

Non-reactive when stored in original container under normal conditions of storage and use.

10.2 Chemical stability

Stable when stored in original container under normal conditions of storage and use.

10.3 Possibility of hazardous reactions

No hazardous reactions when stored in original container under normal conditions of storage and use. May react with strong bases and strong oxidising substances.

10.4 Conditions to avoid

Do not store in proximity of sources of ignition and direct sunlight.

10.5 Incompatible materials
Avoid contact with strong bases and strong oxidising substances.

10.6 Hazardous decomposition products
During decomposition evolves toxic fumes; nitrogen oxides and chlorine oxides.

11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects
11.1.2 Mixtures

All the data contained in this section are derived from actual test data on the mixture unless otherwise stated.

a) Acute Toxicity:

LD_{50} oral, rat:	>2000 mg/kg bw
LD_{50} dermal, rabbit:	>2000 mg/kg bw
LC_{50} inhalation, rat:	Not classified as hazardous under Regulation (EC) 1272/2008
b) Skin Corrosion/Irritation:	not classified as a skin irritant under Regulation (EC) 1272/2008
c) Serious Eye Damage/Irritation:	Not classified as an eye irritant under Regulation (EC) 1272/2008
d) Respiratory or Skin Sensitization:	Not classified as a skin sensitizer in animal studies. No data available for respiratory sensitization
e) Germ Cell Mutagenicity:	Not classified as mutagenic on the basis of mixture component information
f) Carcinogenicity:	Not classified as carcinogenic on the basis of mixture component information
g) Reproductive Toxicity:	Not classified as a reproductive toxicant on the basis of mixture component information
h) STOT-Single Exposure:	Not classified as hazardous for single dose toxicity on the basis of mixture component information
i) STOT-Repeated Exposure:	Not classified as hazardous for repeated dose toxicity on the basis of mixture component information
j) Aspiration Hazard:	Not classified as hazardous by aspiration on the basis of mixture component information

Likely routes of exposure and related long and short term symptoms and health effects:

Inhalation: There is a low risk of exposure by inhalation.

Short-term symptoms and effects:	Long-term symptoms and effects:
Possible slight nasal irritation and discharge.	No evidence of long-term effects after prolonged or repeated exposure.

Eye contact: There is a risk of exposure by eye contact.

Short-term symptoms and effects:	Long-term symptoms and effects:
Possible slight transitory redness.	No evidence of long-term effects after prolonged or repeated exposure.

Skin contact: There is a risk of exposure by skin contact.

Short-term symptoms and effects:	Long-term symptoms and effects:
Possible slight transitory redness.	No evidence of long-term effects after prolonged or repeated exposure.

Ingestion: There is a very low risk of accidental exposure by ingestion.

Short-term symptoms and effects:	Long-term symptoms and effects:
Possible mild gastrointestinal effects.	No evidence of long-term effects after prolonged or repeated exposure.

12. ECOLOGICAL INFORMATION

All the information and data contained in this section are derived from actual test data on the mixture unless otherwise stated.

12.1 Toxicity

Acute Toxicity

LC_{50} fish, *Oncorhynchus mykiss* (96h):	339.8 mg/l
EC_{50} aquatic invertebrates, *Daphnia magna* (48h):	142.7 mg/l
ErC_{50} algae, *Pseudokirchneriella subcapitata* (72h):	395.2 mg/l

EyC_{50} algae, *Pseudokirchneriella subcapitata* (72h):	142.6 mg/l
LD_{50} birds, *Mallard duck:*	Clomazone: >2510 mg/kg
LD_{50} honey bees oral, *Apis mellifera* (48h):	100 µg a.s./bee
LD_{50} honey bees, contact, *Apis mellifera* (48h):	100 µg a.s./bee

Chronic Toxicity

NOEC fish, *Oncorhynchus mykiss* (96h):	43 mg/l
NOEC aquatic invertebrates, *Daphnia magna* (48h):	19 mg/l
NOErC algae, *Pseudokirchneriella subcapitata* (72h):	32 mg/l
NOEyC algae, *Pseudokirchneriella subcapitata* (72h):	10 mg/l

12.2 Persistence and degradability:	Clomazone: Half-life in soil: 77 d (mean value) Not readily biodegradable
12.3 Bioaccumulative potential:	No significant potential for bioaccumulation
12.4 Mobility in soil:	Low to medium mobility in soil
12.5 Results of PBT and vPvB assessment:	No PBT or vPvB assessments have been carried out on the mixture; please refer to 12.1, 12.2 & 12.3.
12.6 Other adverse effects:	Not determined.

13. DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Disposal of waste product, contaminated packaging materials and any excess diluted spray should be in accordance with all relevant national legislation.
For the handling and management of accidental release, follow the information given under Section 6 and 7.

14. TRANSPORT INFORMATION

Not classified as hazardous for transport.

14.1 UN number	-
14.2 UN Proper shipping name	-
14.3 Transport hazard class(es)	-
14.4 Packing group	-
14.5 Environmental hazards	-
14.6 Special Precautions for User	-
14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code	-

15. REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulations

REGULATION (EC) No 1107/2009 of The European Parliament and of the Council of 21 October 2009 concerning the placing of plant protection products on the market and repealing Council Directives 79/117/EEC and 91/414/EEC.
REGULATION (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006.
REGULATION (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC, including amendments.
COMMISSION REGULATION (EU) No 2015/830 of 28 May 2015 amending Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

National Regulations/legislation:

Refer to applicable national classification, packaging and labelling legislation.

15.2 Chemical Safety Assessment
No Chemical Safety Assessment under Regulation (EC) 1907/2006 is required and has not been carried out.

16. OTHER INFORMATION

a) Indication of changes:
The numbering system identifying new versions and/or revisions of this SDS is incremental. An increment by an integer number identifies the issue of a new version requiring provision of updates according to Article 31(9) of REACH, while an increment by a decimal identifies minor changes such as typographical errors, text improvements and/or formatting.
Revisions indicated by a decimal point do not affect the risk management measures or information on hazards, do not refer to restrictions imposed and/or to authorisations granted or refused.
The paragraphs where changes have been made are indicated by the symbol '!' in the margin.

Differences between this version and the previous one: This is the first version of this SDS

b) Abbreviations and acronyms:
Acute Tox. 4: Acute toxicity Category 4
Eye Irrit. 2: Eye Irritation Category 2
Aquatic Acute 1: Hazardous to the Aquatic environment: Acute hazard Category 1
Aquatic Chronic 1: Hazardous to the Aquatic environment: Chronic hazard Category 1
Aquatic Chronic 3: Hazardous to the Aquatic environment: Chronic hazard Category 3
STOT: Specific Target Organ Toxicity

c) Key literature references and sources for data:
Albaugh Europe Sàrl.
ECHA Guidance on the application of the CLP criteria
ECHA Guidance on the compilation of safety data sheets

d) Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]

Classification according to Regulation (EC) no. 1272/2008	Classification method
Aquatic Chronic 3 – H412	Calculation from endpoints and study results

e) Relevant H-statements and precautionary statements not written out in full under Sections 2 to 15:
H302 Harmful if swallowed
H319 Causes serious eye irritation
H332 Harmful if inhaled
H400 Very toxic to aquatic life
H410 Very toxic to aquatic life with long lasting effects.

f) Training advice:
General occupational hygiene training recommended.

g) Further information:
The information and recommendations in this publication are, to the best of our knowledge, information and belief, accurate at the date of publication. Nothing herein is to be construed as a warranty, expressed or implied. In all cases it is the responsibility of the user to determine the applicability of such information or the suitability of any products for their own particular purpose.
This Material Safety Data Sheet was compiled by Albaugh Europe Sàrl (sds@albaugh.eu) in compliance with Regulation (EC) 1907/2006 as amended by 2015/830.